

The conjectured generating function of OEIS A083178, from the recurrence the entry states

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Abstract

OEIS A083178 states a linear recurrence as a fact and carries a rational generating function as an empirical conjecture. The conjecture follows from the recurrence by polynomial algebra alone: the recurrence fixes the denominator, the published terms fix the numerator exactly, and the generating function is then determined as a rational function rather than checked on finitely many coefficients. No model of the sequence and no reading of its name is required.

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1 What the entry says

The first terms are 1, 2, 3, 22, 32, 33, 322, 332, $\dots$, with offset 1. The entry states, as a fact,

For $n > 7$, $a(n) = 11*a(n-3)-10*a(n-6)$. For $n > 4$, $a(n-3) + 3*10^{\lfloor (n-1)/3 \rfloor}$.
For $n > 1$, $(2*10^{\lfloor (n+2)/3 \rfloor} + (63*m^2 - 129*m - 2))/6$, where m is the least nonnegative residue of $n \bmod 3$. - _Chai Wah Wu_, Dec 11 2015

and separately records the conjecture

Empirical g.f.: $x*(90*x^6+10*x^4+11*x^3+3*x^2+2*x+1) / ((x-1)*(x^2+x+1)*(10*x^3-1))$. - _Colin Barker_, Oct 14 2014

The stated line was taken from the entry's factual formulas only. A formula qualified by a finite range – "holds at least up to $n = 1000$ but is not known to hold in general" – is not a fact and is not used, nor is anything inside a conjectural block.

2 The argument

Write $b_k = a(1+k)$ and

$$D(x) = 1 - \sum_i c_i x^i = 10x^6 - 11x^3 + 1,$$

the polynomial of the stated recurrence, of degree 6. Let $B(x) = \sum_{k \geq 0} b_k x^k$.

The recurrence says $b_k = \sum_i c_i b_{k-i}$ for every $k \geq 7$. The coefficient of x^k in $D(x)B(x)$ is exactly $b_k - \sum_i c_i b_{k-i}$, so it vanishes for every $k \geq 7$ and

$$D(x)B(x) = P(x) = \sum_{k < 7} \left(b_k - \sum_i c_i b_{k-i} \right) x^k = 90x^7 + 10x^5 + 11x^4 + 3x^3 + 2x^2 + x,$$

a polynomial whose every coefficient is fixed by finitely many published terms. Therefore

$$B(x) = \frac{P(x)}{D(x)} = \frac{x(90x^6 + 10x^4 + 11x^3 + 3x^2 + 2x + 1)}{(x-1)(10x^3-1)(x^2+x+1)},$$

which is the conjectured expression.

Theorem 1. *The conjectured generating function of OEIS A083178 is correct, given the recurrence the entry states.*

Proof. Immediate from the displayed identity: $DB = P$ holds coefficientwise, $D(0) = 1 \neq 0$ so D is invertible in the ring of formal power series, and $B = P/D$ follows. This is an identity of rational functions, not a comparison of the 30 terms the entry publishes – those enter only in fixing the 7 coefficients of P . $\square$